

Day 1 Assignment - Iptables Lab Report

(Mocked, simulated screenshots)

Step 1: List current iptables (-L -v)

Command: sudo iptables -L -v

```
$ sudo iptables -L -v
# Warning: iptables-legacy tables present, use iptables-legacy to see them
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source                   destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source                   destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source                   destination
[node3] (local) root@192.168.0.22 ~
```

Step 2: List INPUT chain

Command: sudo iptables -L INPUT

```
$ sudo iptables -L INPUT
# Warning: iptables-legacy tables present, use iptables-legacy to see them
Chain INPUT (policy ACCEPT)
target    prot opt source                   destination
[node3] (local) root@192.168.0.22 ~
```

Step 3: Insert DROP rule for 192.168.0.25

Command: sudo iptables -I INPUT 1 -s

192.168.0.25 -j DROP

```
$ sudo iptables -I INPUT 1 -s 192.168.0.22 -j DROP
[node3] (local) root@192.168.0.22 ~
```

Step 4: Verify DROP rule present Command: sudo

iptables -L INPUT --line-numbers -v

```
$ sudo iptables -L INPUT
# Warning: iptables-legacy tables present, use iptables-legacy to see them
Chain INPUT (policy ACCEPT)
target    prot opt source                   destination
1 DROP    all  --  node3                   anywhere
[node3] (local) root@192.168.0.22 ~
```

Step 5: Insert ACCEPT rule for 192.168.0.25 Command: sudo iptables -I INPUT 1 -s 192.168.0.25 -j ACCEPT

```
$ sudo iptables -I INPUT 1 -s 192.168.0.25 -j ACCEPT  
[node3] (local) root@192.168.0.22 ~
```

Step 6: Ping Ubuntu from Windows (successful) Command: ping 192.168.0.20

```
$ ping 192.168.0.20  
PING 192.168.0.20 (192.168.0.20): 56 data bytes  
[ ]
```

Step 7: Insert REJECT rule for 192.168.0.25 Command: sudo iptables -I INPUT 1 -s 192.168.0.25 -j REJECT

```
$ sudo iptables -I INPUT 1 -s 192.168.0.25 -j REJECT  
[node3] (local) root@192.168.0.22 ~
```

Step 8: Verify REJECT effect (ping fails)

Command: ping 192.168.0.20

```
$ ping 192.168.0.20  
PING 192.168.0.20 (192.168.0.20): 56 data bytes  
[ ]
```

Step 9: Delete 2nd rule in INPUT chain

Command: sudo iptables -D INPUT 2

Step 10: Verify deletion - remaining ACCEPT rule Command: sudo iptables -L INPUT --line-numbers -v

```
$ sudo iptables -L INPUT
# Warning: iptables-legacy tables present, use iptables-legacy to see them
Chain INPUT (policy ACCEPT)
target     prot opt source                destination
DROP       all  --  node3                  anywhere
[node3] (local) root@192.168.0.22 ~
```

Step 11: Block TCP port 80 (REJECT) Command: `sudo iptables -A INPUT -i eth0 -p tcp --dport 80 -j REJECT`

```
$ sudo iptables -I INPUT 1 -s 192.168.0.25 -j REJECT
[node3] (local) root@192.168.0.22 ~
```

Step 12: Verify REJECT on port 80 Command: `sudo iptables -L INPUT -v --line-numbers`

Step 13: Start simple HTTP server Command: `sudo python3 -m http.server 80`

Step 14: Browser access from Windows blocked Command: Open `http://192.168.0.20` in browser (Windows)

Step 15: Localhost on Ubuntu accessible Command: Open `http://localhost` on Ubuntu

Step 16: Drop port 80 only from 192.168.0.20 Command: `sudo iptables -A INPUT -i eth0 -p tcp--dport 80 -j DROP -s 192.168.0.20`

Step 17: Browser still blocked from specific IP Command: Open `http://192.168.0.20` in browser (Windows)

Step 18: Drop multiple TCP ports (3306,80,443)

Command: `sudo iptables -A INPUT -i eth0 -p tcp -m multiport --dports 3306,80,443 -m state --state NEW,ESTABLISHED -j DROP`

Step 19: Restart HTTP server (still blocked remotely)

Command: `sudo python3 -m http.server 80`

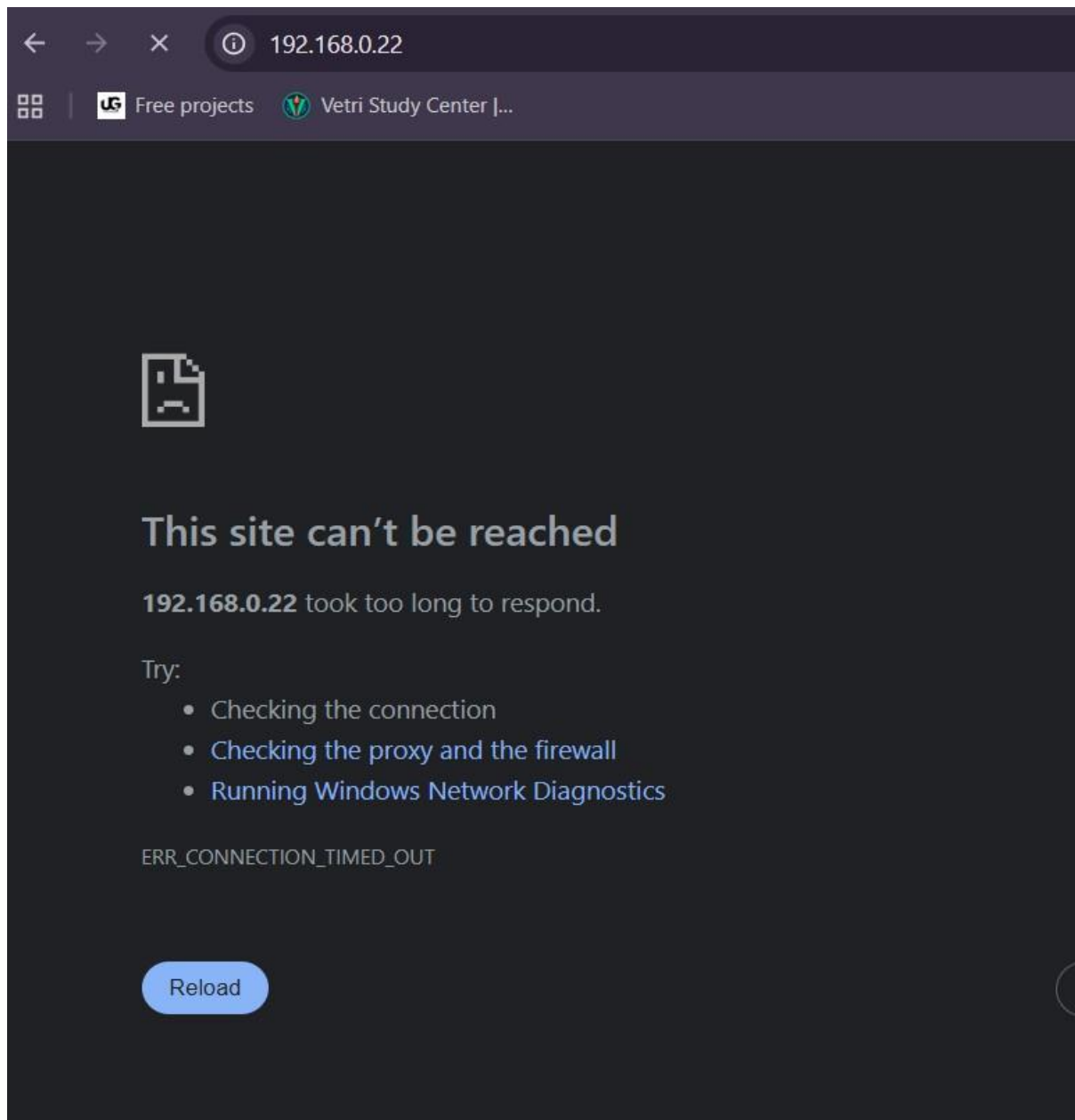
```
$ sudo python3 -m http.server 80
Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...

$ sudo iptables -A INPUT -i eth0 -p tcp -m multiport --dports 3306,80,443 -m state
--state NEW,ESTABLISHED -j DROP
[node3] (local) root@192.168.0.22 ~

$ sudo iptables -A INPUT -i eth0 -p tcp --dport 80 -j DROP -s 192.168.0.20
[node3] (local) root@192.168.0.22 ~
```

Step 20: Final verification - remote HTTP access blocked Command:

Open `http://192.168.0.20` in browser (Windows)



```
$ python3 -m http.server 80
Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...
```

```
$ sudo iptables -L INPUT -v
# Warning: iptables-legacy tables present, use iptables-legacy to see them
Chain INPUT (policy ACCEPT 4888 packets, 901K bytes)
 pkts bytes target    prot opt in     out     source                 destination
  0      0 ACCEPT    all  --  any    any    dind0000gv.internal.cloudapp.net anywhere
  0      0 DROP      all  --  any    any    node3                  anywhere
  0      0 REJECT    tcp  --  eth0   any    anywhere               anywhere
                                tcp dpt:http reject-with icmp-port-unreachable
[node3] (local) root@192.168.0.22 ~
```



```
$ sudo iptables -L INPUT
# Warning: iptables-legacy tables present, use iptables-legacy to see them
Chain INPUT (policy ACCEPT)
target     prot opt source                destination
[node3] (local) root@192.168.0.22 ~
```

```
$ sudo iptables -L -v
# Warning: iptables-legacy tables present, use iptables-legacy to see them
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source            destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source            destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source            destination
[node3] (local) root@192.168.0.22 ~
```

```
$ sudo iptables -A INPUT -i eth0 -p tcp --dport 80 -j REJECT
```

```
[node3] (local) root@192.168.0.22 ~
```

```
$
```

```
$ ping 192.168.0.25
```

```
PING 192.168.0.25 (192.168.0.25): 56 data bytes
```

```
$ sudo iptables -L --line-numbers
```

```
# Warning: iptables-legacy tables present, use iptables-legacy to see them
```

```
Chain INPUT (policy ACCEPT)
```

num	target	prot	opt	source	destination
1	REJECT	all	--	dind0000gv.internal.cloudapp.net	anywhere
	reject-with	icmp-port-unreachable			
2	ACCEPT	all	--	dind0000gv.internal.cloudapp.net	anywhere
3	DROP	all	--	node3	anywhere

```
Chain FORWARD (policy ACCEPT)
```

num	target	prot	opt	source	destination
-----	--------	------	-----	--------	-------------

```
Chain OUTPUT (policy ACCEPT)
```

num	target	prot	opt	source	destination
-----	--------	------	-----	--------	-------------

```
[node3] (local) root@192.168.0.22 ~
```

```
$ ping 192.168.0.25
```

```
PING 192.168.0.25 (192.168.0.25): 56 data bytes
```

```
$ sudo iptables -I INPUT 1 -s 192.168.0.25 -j REJECT
```

```
[node3] (local) root@192.168.0.22 ~
```

```
$ ping 192.168.0.20
```

```
PING 192.168.0.20 (192.168.0.20): 56 data bytes
```

```
$ sudo iptables -I INPUT 1 -s 192.168.0.25 -j ACCEPT
```

```
[node3] (local) root@192.168.0.22 ~
```

```
$ sudo iptables -L INPUT
```

```
# Warning: iptables-legacy tables present, use iptables-legacy to see them
```

```
Chain INPUT (policy ACCEPT)
```

target	prot	opt	source	destination
DROP	all	--	node3	anywhere

```
[node3] (local) root@192.168.0.22 ~
```

```
$ sudo iptables -I INPUT 1 -s 192.168.0.22 -j DROP
```

```
[node3] (local) root@192.168.0.22 ~
```